
MAT1202

Numerical Analysis I

CHAPTER 04 :

Numerical Differentiation and Integration

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Numerical Differentiation

Introduction

Definition:

- Numerical differentiation is the process of approximating the derivative of a function based its values at discrete points.
- This technique is practicaly useful when the function is known only at some specific points or when its analytical differentiation is difficult or impossible.

Importance:

- Numerical differentiation is critical in various scientific and engineering fields where precise measurements exists only at discrete intervals.

Cont.

Applications:

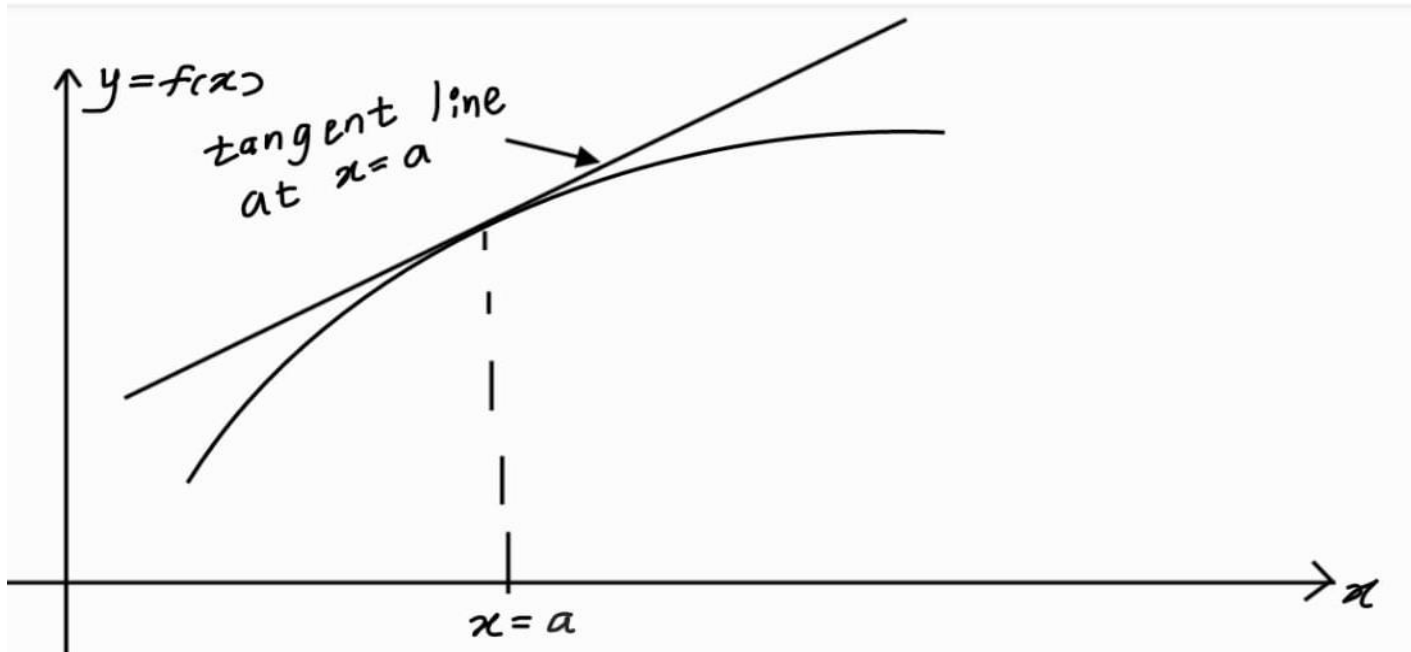
- **Engineering:** Analyzing stress-strain relationships in materials.
- **Physics:** Determining velocity and acceleration from position data.
- **Economics:** Estimating marginal cost and revenue functions.
- **Biology:** Modeling Population dynamics and growth rates.

Cont.

Finite Difference Approximation of derivative:

The derivative $f'(x)$ of a function at the points $x = a$ is defined by

$$\begin{aligned}\frac{dy}{dx} \Big|_{x=a} = f'(a) &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}\end{aligned}$$



$f'(a)$ = the value of the slope of the tangent line to $y = f(x)$ at $x = a$.

Cont.

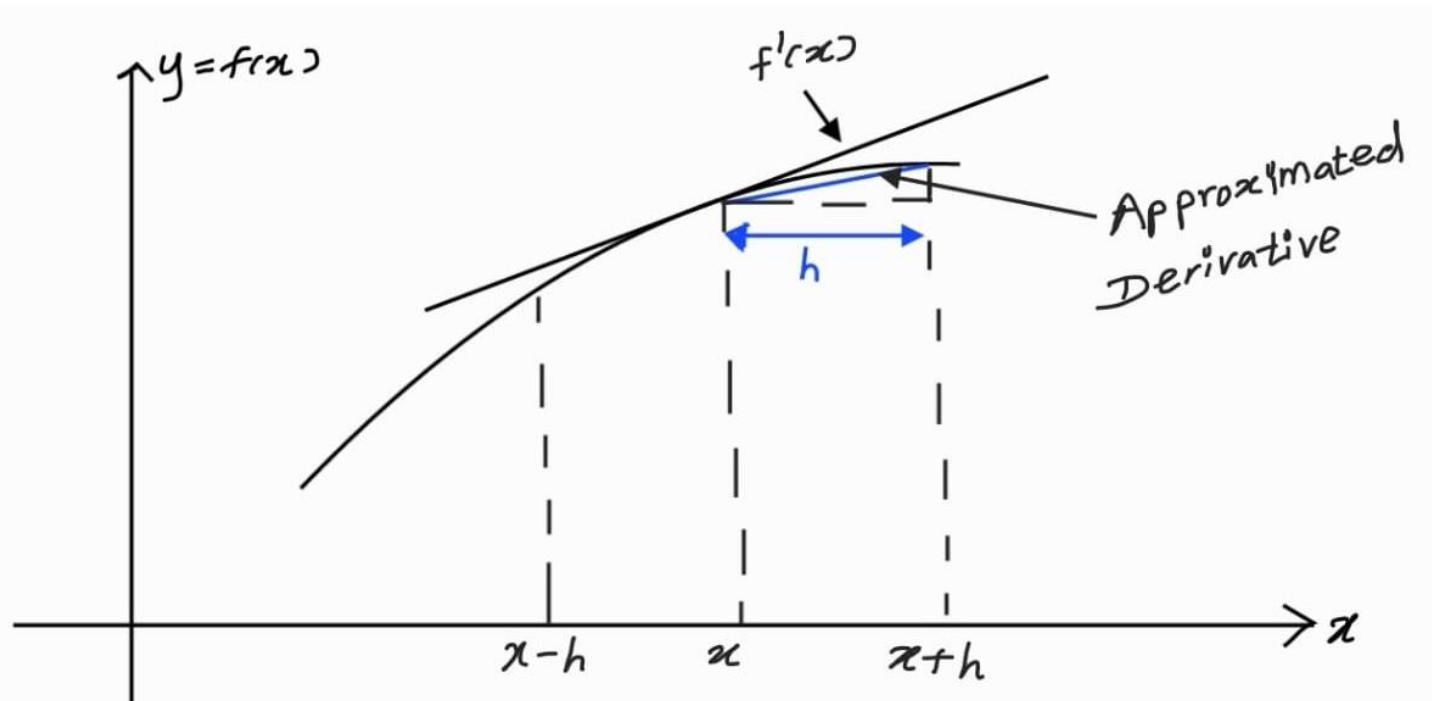
- Finite difference approximations of $f'(a)$ use values of the function $f(x)$ at discrete points in neighborhood of $x = a$, for approximating (estimating) the slope.
- There are various finite difference formulas.

Cont.

1. Forward difference

- Forward difference is the slope of the line that joins points $(x, f(x))$ and $(x + h, f(x + h))$.

$$f'(x) \approx \frac{f(x + h) - f(x)}{h}$$

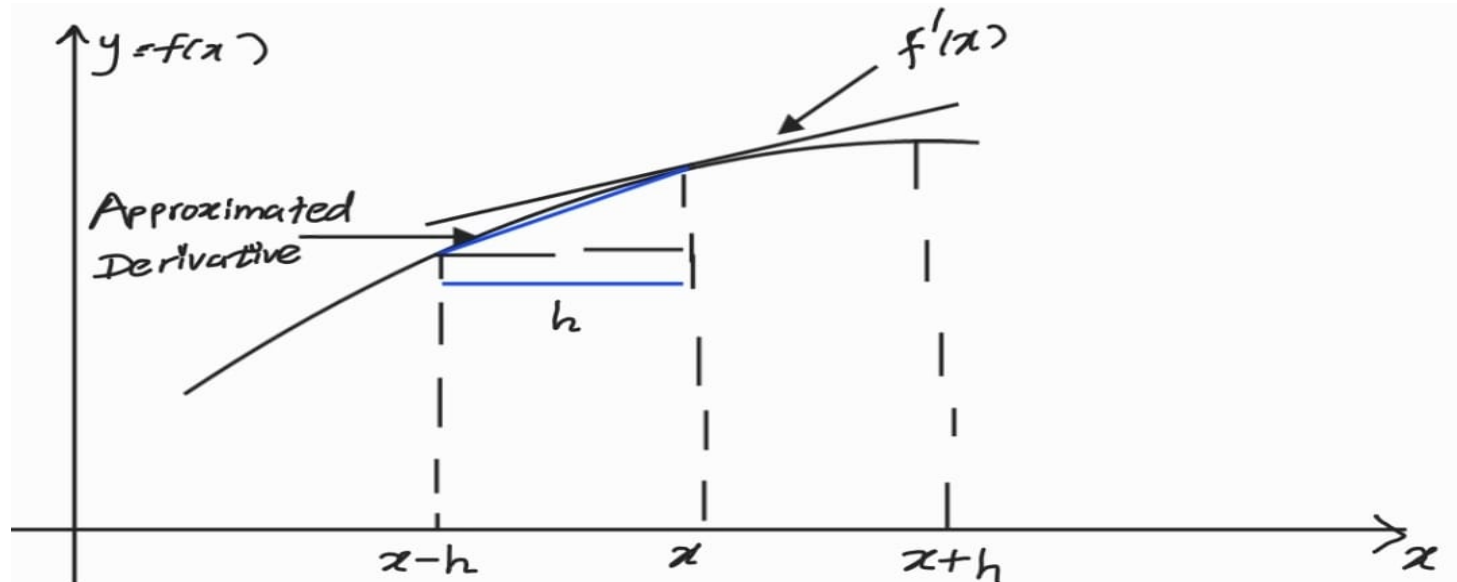


Cont.

2. Backward Difference

- Backward difference is the slope of the line that joins points $(x - h, f(x - h))$ and $(x, f(x))$.

$$f'(x) \approx \frac{f(x) - f(x - h)}{h}$$

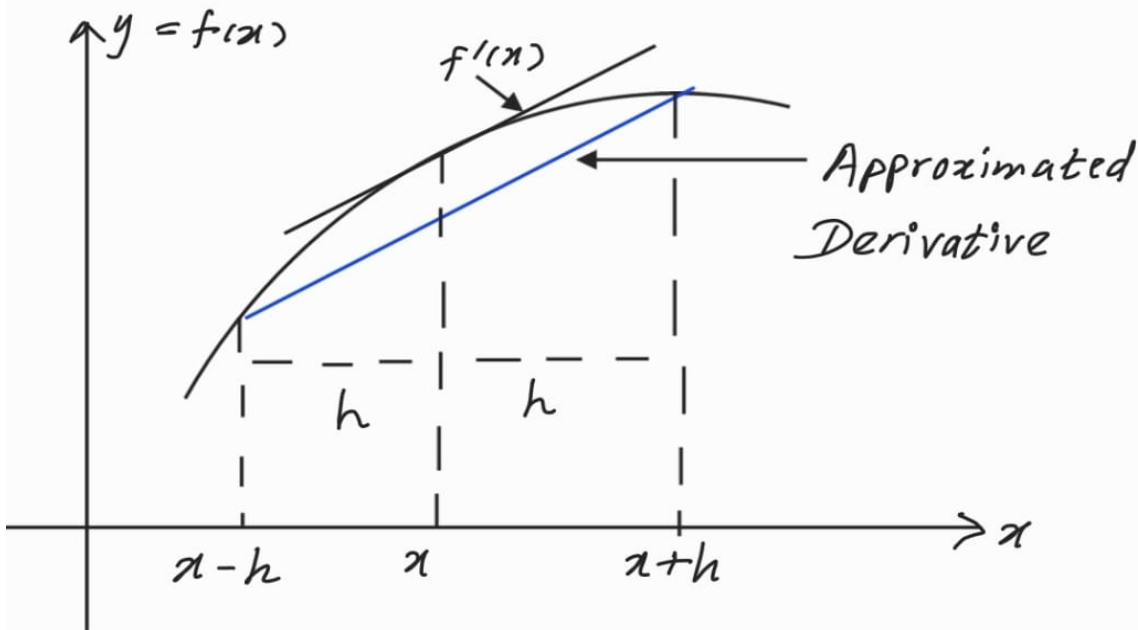


Cont.

3. Central Difference

- Central difference is the slope of the line joining points $(x - h, f(x - h))$ and $(x + h, f(x + h))$.

$$f'(x) \approx \frac{f(x + h) - f(x - h)}{2h}$$



- All the above approximation use only two neighboring points.

Cont.

Example 1:

Consider the function $f(x) = x^2$. Calculate $f'(x)$ at $x = 1$ using the above finite difference approximations. Use $h = 0.25$.

1) Using the forward difference approximation

$$\begin{aligned} f'(1) &\approx \frac{f(1+h) - f(1)}{h} \\ &= \frac{f(1+0.25) - f(1)}{0.25} = \frac{f(1.25) - f(1)}{0.25} = \frac{1.25^2 - 1^2}{0.25} \\ &= 2.25 \end{aligned}$$

The exact solution, $f'(1) = 2 \times 1 = 2$

The absolute relative error = $\left| \frac{2 - 2.25}{2} \times 100\% \right| = 12.5\%$

Cont.

2) Using the backward difference

$$f'(1) \approx \frac{f(1) - f(1 - 0.25)}{0.25} = \frac{f(1) - f(0.75)}{0.25} = \frac{1^2 - 0.75^2}{0.25} = 1.75$$

$$\text{The absolute relative error} = \left| \frac{1.75 - 2}{0.25} \times 100\% \right| = 12.5\%$$

3) Using the central difference

$$f'(1) \approx \frac{f(1 + 0.25) - f(1 - 0.25)}{2 \times 0.25} = \frac{f(1.25) - f(0.75)}{0.5} = \frac{1.25^2 - 0.75^2}{0.5} = 2$$

$$\text{The absolute relative error} = \left| \frac{2 - 2}{2} \times 100\% \right| = 0.00\%$$

Cont.

<H.M>

Consider the function $f(x) = x^3$. Using the forward, backward, and central differences, approximate $f'(3)$. Use $h = 0.25$.

Also, repeat calculation for $h = 0.125, 0.0625, 0.0313$

Finite Difference Formulas using Taylor Series

Taylor Theorem

Let f be a function with $(n + 1)$ derivative $f^{(n+1)}$ continuous on the closed interval $[a, x]$.

Then, for $x \in [a, x]$, $f(x)$ can be written as $f(x) = P_n(x) + R_n(x)$,

where $P_n(x)$ is the n^{th} degree Taylor Polynomial of f centered at a :

$$P_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n$$

and

$$R_n(x) = \frac{f^{(n+1)}(\alpha)}{(n+1)!}(x - a)^{n+1} \text{ for some } \alpha \in (a, x).$$

Cont.

Taylor Series

Let f be a real-valued function. Suppose that $f^{(m)}(a)$ exists for $m = 0, 1, 2, \dots$.

Then $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!} (x - a)^2 + \dots$.

This is called the Taylor Series of f centered at $x = a$.

If $\lim_{n \rightarrow \infty} R_n(x) = 0$, we have

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2} (x - a)^2 + \dots$$

Let $h = x - a$. Then $x = a + h$.

Now,

$$f(a + h) = f(a) + f'(a)h + \frac{f''(a)}{2!} h^2 + \dots$$

Cont.

1. Forward Difference

Consider the Taylor theorem, with $x = a + h$, and $n = 1$.

Then, we have:

$$\begin{aligned} f(a + h) &= f(a) + f'(a)h + R_1(x) \\ &= f(a) + f'(a)h + \frac{f''(\alpha)}{2!} h^2 \\ \Rightarrow f'(a) &= \frac{f(a + h) - f(a)}{h} - \frac{f''(\alpha)}{2!} h \rightarrow \mathbf{1} \end{aligned}$$

- $f'(a)$ can be approximated by $\frac{f(a+h)-f(a)}{h}$ if ignore the term $\frac{f''(\alpha)}{2!} h$ in equation 1.
- Ignoring the term $\frac{f''(\alpha)}{2!} h$ introduces a truncation (discretization) error.
- Since the error $\frac{f''(\alpha)}{2!} h$ is proportional to h , the truncation error is said to be of order 1 which is written as $O(h)$.

Cont.

i.e. ; Truncation error = $\frac{f''(\alpha)}{2!}h = O(h)$.

Now, equation 1 $\Rightarrow f'(a) = \frac{f(a+h)-f(a)}{h} + O(h)$.

- Note that the truncation error $O(h) = \frac{f''(\alpha)}{2!}h$ is not known as α is unknown.

2.Backward Difference

Consider Taylor Theorem with $x = a - h$ and $n = 1$.

$$\begin{aligned}f(a - h) &= f(a) - f'(a)h + \frac{f''(\alpha)h^2}{2!} \\ \Rightarrow f(a - h) &= f(a) - f'(a)h + O(h^2) \\ \Rightarrow f'(a) &= \frac{f(a) - f(a - h)}{h} + O(h)\end{aligned}$$

Cont.

3. Central Difference

Write Taylor Theorem with $n = 2$.

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3$$

$$\Rightarrow f(a + h) = f(a) + f'(a)h + \frac{f''(a)}{2!}h^2 + \frac{f'''(\alpha_1)}{6}h^3 \rightarrow \mathbf{2}; \text{ for } a < \alpha_1 < a + h$$

$$f(a - h) = f(a) - f'(a)h + \frac{f''(a)}{2!}h^2 - \frac{f'''(\alpha_2)}{6}h^3 \rightarrow \mathbf{3}; \text{ for } a - h < \alpha_2 < a$$

$$\mathbf{2} - \mathbf{3} \Rightarrow f(a + h) - f(a - h) = 2f'(a)h + \frac{[f'''(\alpha_1) + f'''(\alpha_2)]}{6}h^3$$

$$\Rightarrow f'(a) = \frac{f(a + h) - f(a - h)}{2h} + \frac{[f'''(\alpha_1) + f'''(\alpha_2)]}{12}h^2$$

$$\Rightarrow f'(a) = \frac{f(a + h) - f(a - h)}{2h} + O(h^2).$$

Truncation error = $O(h^2)$ is of order 2.

Cont.

4. Finite difference approximations for the second derivative

Consider Taylor Theorem with $n = 3$. Then we have:

$$f(a + h) = f(a) + f'(a)h + \frac{f''(a)}{2!}h^2 + \frac{f'''(a)}{3!}h^3 + \frac{f^{(4)}(\alpha_1)}{4!}h^4 \rightarrow \mathbf{4};$$

where $a < \alpha_1 < a + h$

$$f(a - h) = f(a) - f'(a)h + \frac{f''(a)}{2!}h^2 - \frac{f'''(a)}{3!}h^3 + \frac{f^{(4)}(\alpha_2)}{4!}h^4 \rightarrow \mathbf{5};$$

where $a - h < \alpha_2 < a$

$$\mathbf{4} + \mathbf{5} \Rightarrow f(a + h) + f(a - h) = 2f(a) + f''(a)h^2 + \frac{f^{(4)}(\alpha_1) + f^{(4)}(\alpha_2)}{4!}h^4$$

$$f''(a) = \frac{f(a + h) + f(a - h)}{h^2} + O(h^2)$$

Cont.

- This is called the central difference approximation for the second derivative.
- Truncation error = $O(h^2)$ is of order 2.

Example 2:

Let $f(x) = \frac{2^x}{x}$. Approximate $f''(2)$. Use $h = 0.2$ and $h = 0.1$.

When $h = 0.2$

$$\begin{aligned} f''(2) &\approx \frac{f(2 + 0.2) - 2f(2) + f(2 - 0.2)}{(0.2)^2} = \frac{f(2.2) - 2f(2) + f(1.8)}{0.04} \\ &= \frac{2.0885 - 2(2) + 1.9346}{0.04} = \frac{0.0231}{0.04} = 0.5775 \end{aligned}$$

Example

When $h = 0.1$

$$\begin{aligned} f''(2) &\approx \frac{f(2 + 0.1) - 2f(2) + f(2 - 0.1)}{(0.1)^2} = \frac{f(2.1) - 2f(2) + f(1.9)}{0.01} \\ &= \frac{2.0415 - 2(2) + 1.9643}{0.01} = \frac{0.0058}{0.01} = 0.5753 \end{aligned}$$

Analytical differentiation

$$f(x) = \frac{2^x}{x} \Rightarrow f'(x) = \frac{2^x \ln(2)}{x} - \frac{2^x}{x^2}; \text{ (Note: } \frac{d(2^x)}{dx} = 2^x \ln 2 \text{)}$$

$$f''(x) = \frac{2^x (\ln 2)^2}{x} - \frac{2 \cdot 2^x \ln(2)}{x^2} + \frac{2 \cdot 2^x}{x^3}$$

$$f''(2) = 0.5746$$

Cont.

Error Calculation:

$$\text{When } h = 0.2, \text{ error} = \left| \frac{0.5775 - 0.5746}{0.5746} \times 100\% \right| = 0.5047\%$$

$$\text{When } h = 0.1, \text{ error} = \left| \frac{0.5753 - 0.5746}{0.5746} \times 100\% \right| = 0.1218\%$$

Questions/Queries?

